

CYBERSENTINEL

Behavioral Network Intrusion Detection & Response Prototype

Technical Implementation Report

Version 2.0 — February 2026

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1. Introduction

CyberSentinel v2 represents an evolution of the initial behavioral intrusion detection prototype.

While version 1 focused on generic network-level detection mechanisms such as ICMP flood and TCP SYN scanning, version 2 introduces service-aware detection capabilities.

The primary enhancement in this version is the implementation of SSH brute-force detection, enabling the system to identify targeted credential-based attacks in real time.

This upgrade transforms CyberSentinel from a generic threshold-based IDS into a more service-aware intrusion detection and response prototype.

2. Project Evolution

CyberSentinel v2 builds upon the stable architecture developed in version 1.

The following enhancements were implemented:

- Service-specific detection (SSH – Port 22)
- Critical severity classification for credential-based attacks
- Improved threat scoring model
- Enhanced detection granularity
- Maintained modular architecture

This version demonstrates progressive system improvement while preserving architectural stability.

3. Updated System Architecture

CyberSentinel v2 preserves the modular architecture introduced in version 1 while extending its detection capabilities.

The system now includes service-aware detection mechanisms in addition to protocol-level behavioral analysis.

The updated architecture consists of:

1. Packet Capture Module
Captures live traffic using Scapy.

2. Protocol-Level Detection

- ICMP Flood Detection
- Generic TCP SYN Scan Detection

3. Service-Aware Detection

- SSH Brute-Force Detection (Port 22 monitoring)

4. Logging Module

Generates structured JSON alerts.

5. Automated Response Module

Dynamically blocks malicious IP addresses using iptables.

This layered architecture enables CyberSentinel to detect both volumetric attacks and targeted credential-based attacks in real time.

4. SSH Brute-Force Detection Logic

CyberSentinel v2 introduces service-aware detection specifically designed to identify SSH brute-force attacks.

Unlike volumetric attacks such as ICMP floods, SSH brute-force attacks target authentication mechanisms by generating repeated connection attempts within a short time interval.

The detection mechanism implemented in CyberSentinel v2 monitors TCP SYN packets directed toward destination port 22 (SSH).

For each source IP address, connection attempts are tracked within a sliding time window of 5 seconds.

Detection parameters:

- Protocol: TCP
- Destination Port: 22 (SSH)
- Time Window: 5 seconds
- Threshold: 3 connection attempts (for testing environment)
- Severity Level: CRITICAL
- Threat Score: 85

If the number of SSH connection attempts exceeds the defined threshold within the time window, the system classifies the activity as a potential brute-force attack.

Once detected, CyberSentinel automatically:

- Generates a CRITICAL severity alert
- Logs the event in structured JSON format
- Dynamically inserts a firewall rule to block the attacker's IP address

This enhancement transforms CyberSentinel from a generic network-level detector into a service-aware intrusion detection and automated response system.

5. SSH Brute-Force Attack Simulation

To validate the SSH brute-force detection mechanism implemented in CyberSentinel v2, a controlled attack simulation was conducted in a virtual lab environment.

5.1 Test Environment

- Attacker Machine: Kali Linux (VirtualBox)
- Target Machine: Ubuntu Server running CyberSentinel v2
- Service Targeted: SSH (Port 22)
- Detection Engine: Python + Scapy
- Firewall: iptables

5.2 Attack Execution

A brute-force attack was simulated using Hydra, a widely used penetration testing tool capable of performing credential-based attacks against network services.

The following command was executed from the Kali Linux machine:

```
hydra -l fakeuser -P /usr/share/wordlists/rockyou.txt ssh://192.168.0.162 -t 4
```

This command generated multiple rapid SSH authentication attempts against the target system.

```
(ismael_kali@vbox)-[~]
$ for i in {1..5}; do nmap -p 22 -sS -T5 192.168.0.162; done
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-22 16:27 AST
Nmap scan report for 192.168.0.162
Host is up (0.0027s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-22 16:27 AST
Nmap scan report for 192.168.0.162
Host is up (0.0038s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-22 16:27 AST
Nmap scan report for 192.168.0.162
Host is up (0.0038s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-22 16:27 AST
Nmap scan report for 192.168.0.162
Host is up (0.0032s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-22 16:27 AST
Nmap scan report for 192.168.0.162
Host is up (0.0028s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:75:75:36 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
```

5.3 Detection and Response

During the attack, CyberSentinel monitored incoming TCP SYN packets targeting port 22.

Once the number of connection attempts exceeded the defined threshold within the 5-second time window, the system triggered:

- A CRITICAL severity alert
- A structured log entry
- An automatic firewall rule insertion

The system output displayed:

```
ismael@ismael:~/cybersentinel$ ismael@ismael:~/cybersentinel$ sudo python3 main.py
[*] CyberSentinel v2 running on enp0s3
[🚨 ALERT] SSH Brute Force suspected from 192.168.0.135
[ALERT LOGGED] SSH_BRUTE_FORCE from 192.168.0.135
[🛑 BLOCKED] IP 192.168.0.135 has been blocked by firewall
```

5.4 Mitigation Verification

The firewall configuration was verified using:

```
sudo iptables -L
```

The attacker's IP address appeared in the DROP rule list, confirming successful automated mitigation.

After the IP address was blocked, further SSH attempts from the attacker machine were denied.

```
ismael@ismael:~/cybersentinel$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  192.168.0.135         anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ismael@ismael:~/cybersentinel$
```

6. Comparative Analysis (v1 vs v2)

CyberSentinel v2 represents a significant evolution from the initial version.

While version 1 focused primarily on protocol-level behavioral detection (ICMP flood and generic TCP SYN scan detection), version 2 introduces service-aware detection capabilities.

Key Improvements in Version 2

1. Service-Level Monitoring

Version 1 detected volumetric network anomalies.

Version 2 introduces targeted monitoring of SSH (Port 22), enabling detection of credential-based attacks.

2. Critical Severity Classification

SSH brute-force attacks are classified as CRITICAL due to their potential impact on system compromise.

3. Enhanced Threat Modeling

Version 2 differentiates between:

- Network reconnaissance (SYN scan)
- Volumetric flooding (ICMP)
- Credential-based attacks (SSH brute-force)

4. Improved Realism

The use of Hydra for testing aligns CyberSentinel v2 with real-world attack scenarios.

5. Defensive Automation Maturity

Both versions implement automated firewall mitigation, but v2 demonstrates more precise and context-aware blocking.

Summary of Evolution

Version 1: Network-Level Behavioral Detection

Version 2: Service-Aware Detection with Realistic Attack Simulation

This progression demonstrates iterative system improvement, enhanced detection granularity, and stronger defensive capabilities.

7. Future Improvements

While CyberSentinel v2 introduces service-aware detection and automated mitigation, several improvements can further enhance the system's capabilities.

Planned future enhancements include:

1. **Stateful Flow Tracking**
Implementation of full 5-tuple session tracking (source IP, destination IP, source port, destination port, protocol) to enable deeper traffic analysis.
2. **Cumulative Threat Scoring**
Assign dynamic risk scores to IP addresses based on repeated suspicious behaviors.
3. **Multi-Service Monitoring**
Extend detection logic to additional services such as:
 - FTP
 - HTTP
 - SMB
4. **Anomaly-Based Detection**
Introduce statistical baseline modeling to detect deviations from normal traffic patterns.
5. **Web-Based Monitoring Dashboard**
Develop a real-time visualization interface for alerts and traffic statistics.
6. **Persistent Firewall Rules**
Improve mitigation persistence across system reboots.

These enhancements aim to progressively transform CyberSentinel into a more advanced and scalable network intrusion detection and response system.

8. Conclusion

CyberSentinel v2 demonstrates the successful evolution of a behavioral intrusion detection prototype into a service-aware detection and automated response system.

By integrating SSH brute-force detection, the system moves beyond generic network anomaly monitoring and begins addressing realistic attack vectors commonly observed in operational environments.

The project highlights several key competencies:

- Real-time packet inspection using Scapy
- Threshold-based behavioral detection
- Service-specific attack identification
- Structured alert logging
- Automated firewall-based mitigation

The iterative development from version 1 to version 2 reflects a structured engineering approach, emphasizing progressive enhancement, validation through controlled attack simulation, and architectural stability.

CyberSentinel v2 serves as a foundational step toward building a more advanced Network Detection and Response (NDR) platform capable of handling increasingly sophisticated threat scenarios.